AWS Management Console, CLI, and SDK

Explore ways to interact with AWS using the Management Console, Command Line Interface, and SDKs. Understand how to create and manage resources like S3 buckets through graphical, command-line, and programmatic methods.

AWS offers users various ways to interact with services and create, configure, and manage resources. These include AWS Management Console, AWS Command Line Interface (CLI), and Code using Software Development Kits (SDKs).

AWS Management Console

The **AWS Management Console** is a web-based interface that allows users to access and manage their AWS resources and services. When we log in to our AWS account, we land on the AWS Management Console, which allows us to search and navigate the service dashboards. It is a central hub for interacting with various AWS offerings, enabling users to effectively deploy, monitor, and administer cloud-based solutions.

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The current region is displayed in the top-right corner of the console. We can click the region name to open the region selector. The region selector displays a list of AWS regions available. Each region is identified by its name (such as US East (N. Virginia), EU (Ireland), Asia Pacific (Tokyo), and others).

Create an S3 bucket using the AWS Management Console

Amazon Simple Storage Service (S3) is a highly scalable object storage service. In Amazon S3, a bucket is a container for storing objects, ranging from data files to application code. Each bucket has a unique name and is located in a specific AWS region. Let's go through the steps of creating a bucket using the AWS Management Console:

- Open the AWS Management Console and in the search bar, search for "S3" and select "S3" from the search results.
- On the S3 dashboard, click the "Create" button. This will redirect us to the new bucket page.
- Here, enter the name of the bucket `console-bucket`. Note that the name of the bucket must be globally unique.
- Keep the rest of the settings and click "Create bucket."

Thus, we can create an S3 bucket using the AWS Management Console in a few clicks.

AWS Command Line Interface

The **AWS Command Line Interface (CLI)** allows users to interact with AWS services and resources from the command line or scripts. It provides a powerful and efficient way to automate tasks, manage AWS infrastructure, and access AWS services programmatically.

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The AWS CLI is available for Windows, macOS, and Linux. Once the AWS CLI is installed, you must configure your AWS credentials to authenticate with AWS services. You can configure credentials using the command given below:

```
aws configure
```

The command requires entering your AWS Access Key ID, Secret Access Key, default region, and output format.

> **Note:** You can get your AWS Access Key ID and Secret Access Key by navigating to the AWS Management Console. Here, click on your account name and then select “Security Credentials.” Locate the “Access keys” section and click “Create access key.”

Create an S3 bucket using AWS CLI

In the previous section, we created an S3 bucket using the AWS Management Console. We can also achieve the same using the AWS CLI. The command given below creates an S3 bucket `cli-bucket` in the region `us-east-1`.

```
aws s3api create-bucket --bucket cli-bucket --region us-east-1
```

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> **Note:** Remember that the bucket's name should be globally unique. Therefore, if executing the above command gives an error message stating that the bucket already exists, append your account ID or some random characters to the bucket name to make it unique.

We can also add objects to the bucket through the CLI. The command given below will add a file, `my-file.txt`, to the bucket `cli-bucket`.

```
aws s3 cp /my-file.txt s3://cli-bucket/my-file.txt
```

Similarly, we can create, delete, and modify resources for other AWS services using the CLI.

AWS Software Development Kits (SDKs)

AWS has designed **Software Development Kits** (SDKs) that interact with AWS services through code and API calls. The SDKs are available in commonly used languages such as Python, Java, .NET, and more.

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SDKs for languages supported by AWS

AWS SDKs for different programming languages include easy-to-use libraries and API calls, which simplify the development, management, and deployment of applications.

Create an S3 bucket using AWS SDK

With the help of AWS SDKs, we can create multiple AWS resources. The code snippet given below uses the `boto3` library of Python SDK to create an S3 bucket.

The code first launches an S3 client using the `client` function of `boto3`. The client is then used to create a bucket in `us-east-1` region.

main.py Python

```
import boto3

s3_client = boto3.client('s3', region_name='us-east-1')
bucket_name = "sdk-bucket"

try:
    response = s3_client.create_bucket(
        Bucket=bucket_name
    )
    print(f"Bucket created successfully.")
except Exception as e:
    print(f"Error creating bucket: {e}")
```

Python code to create an S3 bucket

To ease out the developers, AWS offers multiple ways to launch, configure, and delete AWS resources. These options give users more control over the resources.

Conclusion

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AWS offers multiple ways to create, manage, and delete AWS resources. Developers can choose among these according to convenience, usability, and

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